

HARSHIT DHANKHAR

San Diego, CA

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Education

University of California San Diego (UCSD)

Sep 2025 – Dec 2026 (Expected)

Master of Science in Computer Science; **GPA: 3.925/4.0**

San Diego, CA

Indian Institute of Technology (IIT) Patna

Jul 2021 – May 2025

Bachelor of Science in Mathematics and Computing; **GPA: 3.75/4.0**

Patna, India

- **Coursework:** Data Structures, Algorithms, Operating Systems, Databases, Software Engineering, Deep Learning, Computer Vision, Data Science, Convex Optimization, Probability Theory, Statistics, Artificial Intelligence

Technical Skills

Languages: Python, C, C++, Java, TypeScript, JavaScript, MATLAB, R, SQL

Technologies: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, RLHF, PEFT (LoRA/QLoRA), FAISS/Pinecone, RAGAS, Pytest, LangChain, LangGraph, MCP, CI/CD, Git, MLFlow, Weights & Biases, Docker, Kubernetes, FastAPI, Grafana, React.js, Redis, Bash, AWS (Bedrock, EC2, S3, Lambda), GCP, Google vertex AI, MongoDB, PostgreSQL, Node.js

Experience

Amazon Web Services (AWS)

Jun 2026 – Sep 2026

Software Development Engineer (SDE) Intern

Seattle, WA

- Engineered an **operations agent** hosted on **OpenClaw** that autonomously generates and delivers pipeline health, dashboard anomaly, and ticket queue reports to Slack, reducing on-call review time from **4+ hours to 30 minutes**.
- Deployed an **auto-incident response system** for Sev-2 tickets using an **RCA agent** that runs a **multi-hypothesis loop** to investigate incidents and posts root cause and customer impact analysis within **15 minutes**.

AnyFeast

Feb 2025 – Aug 2025

AI Engineer

London, UK (Remote)

- Shipped a zero-touch **ETL pipeline** that automates recipe attribute extraction using **agentic AI** and enriches recipe instructions with a **LangGraph-powered RAG module**, reducing manual effort by over **90%**.
- Implemented a **hybrid filtering recommender system**, replacing vanilla popularity-based sorting and **increasing CTR by 30%**, with improvements validated through **A/B testing** and **Azure Monitor**.

LAMSADE, Universite Paris-Dauphine - PSL

May 2024 – Oct 2024

AI Research Intern — [Code], [DOI]

Paris, France

- Developed a novel **Nested Monte Carlo Search (NMCS)**-based algorithm optimizing qubit routing, achieving a **10.55% reduction in circuit depth** versus baselines from IBM (Qiskit) and Google (Cirq).
- Employed mutex-locked qubit scheduling, combinatorial SWAP-set construction, and action-masked payout pruning, enabling sample-efficient exploration and delivering a **37.17% lower average runtime**.
- Presented this **first-author**, peer-reviewed research on quantum circuit optimization at **AAAI 2025 (QC+AI)** in PA.

Projects

Alloc8 - Dorm room Allocation Portal | MongoDB, Express.js, React.js, Node.js

Demo | Website

- **Led a team of 8** developers to deliver Alloc8, a dorm room allocation portal for **3000 students across 1400 rooms** at IIT Patna, **saving an estimated 200+ hours** of administrative and student time every year.
- Deployed backend on **VPS with load balancing** ensuring over 95% uptime during high-traffic; implemented **Redis-based locks** for secure room selection and performed roommate pairing using **Spherical K-Means** clustering.

Deep RL for EV Charging Path Optimization | Python, RL Agents, OpenAI Gym

Code | Paper

- Designed a custom Gym environment for Xi'an city and trained a **PPO-based DRL agent** to optimize EV path routing for charging stations, reducing cumulative driver costs by an average of **74.19%** across baseline routing strategies.
- Published the end-to-end framework in **Elsevier's Sustainable Energy, Grids and Networks journal**.

Selected Publications

Tabular and Deep Reinforcement Learning for Gittins Index | WiOpt 2025

DOI

- Developed novel tabular and deep Q-learning-based algorithms for estimating Gittins indices in **Multi-Armed Bandits (MABs)**, surpassing state-of-the-art methods in memory (**~50%**), runtime (**~10%**), and convergence.

Analysis of Carbon Emission in SAARC Countries using Attention-based LSTM | IEEE BigData 2023

DOI

- Implemented a framework for deploying Deep Learning models (GRU, LSTM, Attn-LSTM, etc) to **forecast CO₂ emissions** in SAARC countries.